SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1710

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REG NO: 192511352

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks: 50

Annexure A

Analytical Problem Solving

Student 25: Preethi R | Register Number: 192511352

1. For a chess-playing agent, characterize the task environment across all six PEAS / environment dimensions (observable, deterministic, episodic, static, discrete, agents). Justify each classification with reference to the rules of chess.

2. Trace the working of Breadth-First Search and Uniform Cost Search on a weighted graph with at least 7 nodes where edge costs are unequal. Show why BFS may not find the optimal (least-cost) path while UCS does, using your example graph.

3. Solve the 6-Queens problem: place 6 queens on a 6x6 chessboard such that no two queens attack each other. Formulate the problem components (state space, initial state, successor function, goal test) and explain how incremental formulation reduces the search space compared to complete-state formulation.

CO- 1 ASSESSMENT TOOL - 1

Analytical problem Solving

Name : R. Preethi
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course code: 1710

1) Chess - playing Agent - Characterization of Task Environment

Introduction:

*A chess - playing agent is an intelligent System that observes the current board position, evaluate possible moves and Selects the best move to achieve checkmate while preventing the opponent from doing the Same.

*The task environment can be analyzed using Six AI environment characteristics.

Observable:

*The chess environment is fully observable because every piece on the board is visible to both players.

* No information is hidden, and each player knowns the exact position of every piece before making a move. Since the agent has complete knowning of the current State, it does not need to estimate unknown, information.

Deterministic:

* Chess is a deterministic environment because every legal move produces a predictable result according to the rules of the game.

*There is no randomness Such as rolling dice or drawing Cards.

* If the Same move is played
position, the outcome will alway be identical.

Episodic or Sequential:

* Chess is a Sequential task because every move influence future position.

* A poor move in the opening may create weakness that affect the middle game and end geme.

* Therefore, each decision must consider both current and future consequences.

Static or Dynamic:

* Chess is generally considered static because the board position does not change while a player is thinking.

* The environment remains unchanged until one of the players makes a move.

Discrete or Continuous:

* Chess is discrete because it has a finite number of board position, piece and legal moves. players make moves one at a time and each move changes the board to another distinct State.

Single - Agent (or) multi- Agent:

* Chess is a multi-Agent environment because two intelligent agents compete against each other.

* The Success of one player depends on predicting and countering the opponent's action

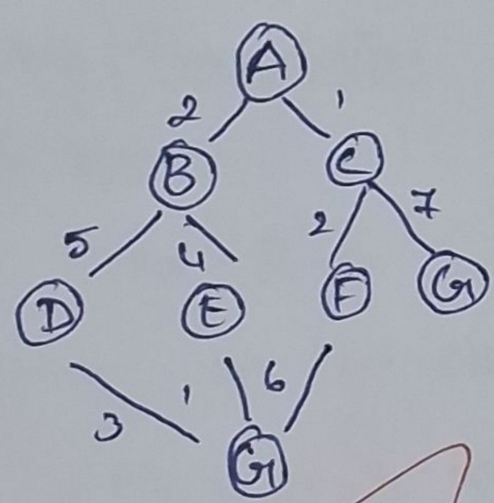
Conclusion :

* Chess represents a fully observable, deterministic Sequential, static, discrete and multi-agent environment.

* These characteristic make chess an ideal problem for artifical intelligence because algorithms must combine Search techniques, heuristics and strategic planning to Select optimal moves.

2. Breadth - first Search (BFS) and uniform cost Search

Problem Statement



Goal : Find the least - cost path from A to G.

Breadth first Search

Edge Cost

$A \rightarrow B = 2$

$A \rightarrow C = 1$

$B \rightarrow D = 5$

$B \rightarrow E = 4$

$C \rightarrow F = 2$

$C \rightarrow G = 7$

$D \rightarrow G = 3$

$E \rightarrow G = 1$

$F \rightarrow G = 6$

Expansion order:

1. A
2. B, C
3. D, E, F, G

First goal found:

$A \rightarrow C \rightarrow G$

cost = 1+7 = 8

Uniform cost Search

Expanded Node	path Cost
A	0
C	1
B	2
F	3
E	6
G	7

optimal path

A → B → E → G

Cost = 2+4+1=7

BFS minimize the number of edges but ignores edges costs. UCS always expands the lowest-cost path first, so it finds the least-cost Solution

6- Queen problem

Problem Statement

* The objective is to place six queens on a 6×6 Chessboard Such that no two queens attacks each other.

* Queens can attack horizontally, vertically and diagonally.

Problem Formulation

State Space:

* The State Space consists of every possible arrangement of queens on the Chessboard.

* Each State represents a partial or complete placement of queens.

Initial State:

* The initial State is an empty 6×6 Chessboard with no queens placed.

Successor Function

* Generate a new State by placing one queen in the next row, ensuring that the Selected column does not conflict with previously placed queens.

Goal Test:

* The goal is reached when six queens are Successfully placed without any conflicts in rows, columns or diagonals.

One valid Solution

Row	column
1	2
2	4
3	6
4	1
5	3
6	5

Board representation

```
.  Q  .  .  .  .
.  .  .  Q  .  .
.  .  .  .  .  Q
Q  .  .  .  .  .
.  .  Q  .  .  .
.  .  .  .  Q  .
```

Complete - State formulation

* In complete - state formulation, all six queens are placed first without checking constraints.

* After generating the complete arrangement the algorithm verifies whether any queens attack each other

Analysis

→ very large Search Space.

→ High Computational cost

Incremental Formulation

* In incremental formulation, queens are placed one row at a time.

* After placing each queen, conflicts are checked immediately.

* If a conflict occurs that branch is abandoned through backtracking.

Conclusion

* The incremental formulation is significantly more efficient because it prunes invalid states during the search process rather than after generating complete board configuration.

* This reduces the search space and computational efforts, making it the preferred approach from solving the 6- Queens problem using backtracking.

Code of Conduct:

I Preethi R (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R. Preethi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

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